# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

**ANSWER KEY & MARKING SCHEME**

## Daily Practice: Ch 4 Carbon and its Compounds

*Chapter / Topic: Carbon and its Compounds · Week 2 of the 90-day plan*

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### DAY 1: BONDING AND STRUCTURES

**Q1.** How many electrons are shared between the two nitrogen atoms in a molecule of $N_2$?

**Answer:** (c) 6: each nitrogen shares three electrons, forming a triple bond (N≡N)

**Q2.** Which form of carbon is made of football-shaped molecules of 60 carbon atoms?

**Answer:** (d) buckminsterfullerene, $C_{60}$

**Q3.** The molecular formula of cyclohexane is:

**Answer:** (b) $C_6H_{12}$: a ring of six carbons, each with two hydrogens

**Q4.** The general formula of the alkynes is:

**Answer:** (c) $C_nH_{2n-2}$, e.g. ethyne $C_2H_2$, propyne $C_3H_4$

**Q5.** Draw the electron-dot structures of (a) an oxygen molecule, $O_2$, and (b) carbon dioxide, $CO_2$.

**Answer:** (a) The two O atoms share two pairs of electrons: a double bond, O=O. Each O also keeps two lone pairs.
(b) Carbon shares two pairs with each oxygen: O=C=O, two double bonds. Each O keeps two lone pairs.

**Q6.** Diamond and graphite are both pure carbon. Why is diamond very hard, while graphite is soft and slippery?

**Answer:** In diamond each carbon is bonded to four others in a rigid three-dimensional network, so it is very hard. In graphite each carbon is bonded to three others in flat hexagonal layers; the layers slide over one another, so graphite is soft and slippery.

**Q7.** Write the structural formula and the molecular formula of (i) propane, (ii) propene and (iii) propyne.

**Answer:** (i) Propane: $CH_3–CH_2–CH_3$, $C_3H_8$
(ii) Propene: $CH_3–CH=CH_2$, $C_3H_6$
(iii) Propyne: $CH_3–C≡CH$, $C_3H_4$

**Q8.** Draw the structures of cyclohexane and benzene. How is the bonding different?

**Answer:** Cyclohexane, $C_6H_{12}$: a ring of six carbons joined by single bonds, each carbon with two H atoms.
Benzene, $C_6H_6$: a ring of six carbons with alternate single and double bonds, each carbon with one H atom.
Cyclohexane is saturated; benzene has double bonds, so it is unsaturated.

### DAY 2: FUNCTIONAL GROUPS, NAMING AND ISOMERS

**Q9.** The IUPAC name of $CH_3–CH_2–CH_2–COOH$ is:

**Answer:** (d) butanoic acid: four carbons, with the –COOH group

**Q10.** Which pair belongs to the same homologous series?

**Answer:** (b) $C_2H_6$ and $C_3H_8$: both alkanes, differing by $–CH_2–$

**Q11.** The formula of chloropropane is:

**Answer:** (a) $C_3H_7Cl$: one H of propane ($C_3H_8$) is replaced by Cl

**Q12.** Name the functional group in each: (a) $CH_3CHO$ (b) $CH_3COCH_2CH_3$ (c) $CH_3CH_2COOH$ (d) $CH_3Cl$.

**Answer:** (a) aldehyde (–CHO) (b) ketone (C=O inside the chain) (c) carboxylic acid (–COOH) (d) halo group (chloro, –Cl)

**Q13.** Draw the structures of (a) ethanoic acid, (b) bromopentane and (c) butanone.

**Answer:** (a) $CH_3$–COOH, i.e. $CH_3$–C(=O)–OH
(b) $CH_3$–$CH_2$–$CH_2$–$CH_2$–$CH_2$–Br
(c) $CH_3$–CO–$CH_2$–$CH_3$, i.e. $CH_3$–C(=O)–$CH_2$–$CH_3$

**Q14.** Pentane, $C_5H_{12}$, has three structural isomers. Draw all three.

**Answer:** 1. A straight chain: $CH_3$–$CH_2$–$CH_2$–$CH_2$–$CH_3$
2. A branch on the second carbon: $CH_3$–$CH(CH_3)$–$CH_2$–$CH_3$
3. Two branches on the middle carbon: $C(CH_3)_4$, a central carbon joined to four $CH_3$ groups.
All three have the formula $C_5H_{12}$.

**Q15.** Carbon forms neither $C^{4+}$ nor $C^{4-}$ ions. Explain why, and say how carbon achieves a stable configuration instead.

**Answer:** To form $C^{4+}$, carbon would have to lose four electrons, which needs a very large amount of energy. To form $C^{4-}$, it would have to gain four electrons, and a nucleus with six protons cannot hold ten electrons easily. So carbon shares its four valence electrons with other atoms, forming covalent bonds.

## DAY 3: REACTIONS, ETHANOL AND ETHANOIC ACID

**Q16.** **Assertion (A):** Oils containing unsaturated fatty acids should be chosen for cooking. **Reason (R):** Animal fats generally contain saturated fatty acids,…

**Answer:** (a) Both are true, and R explains A.

**Q17.** Denatured alcohol is ethanol made unfit for drinking by adding:

**Answer:** (a) methanol (a poison) and a blue dye, so that it can be recognised

**Q18.** Which of these reacts with chlorine in sunlight by substitution?

**Answer:** (c) $CH_4$: saturated hydrocarbons give substitution reactions. $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$

**Q19.** A teacher gave Aryan two unlabelled bottles, one of ethanol and one of ethanoic acid. He added a pinch of sodium hydrogencarbonate to a little of…

(i) Ethanoic acid: $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$
(ii) Pass it through lime water: it turns milky, so the gas is carbon dioxide.
(iii) Ethanoic acid turns blue litmus red; ethanol does not. (Ethanoic acid also smells of vinegar.)

**Q20.** (a) Write the balanced equation for the complete combustion of ethanol. (b) What is an oxidising agent? Name two substances that can oxidise ethanol…

**Answer:** (a) $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ + heat and light
(b) A substance that adds oxygen to another substance. Alkaline potassium permanganate and acidified potassium dichromate.
(c) Vinegar is a 5–8% solution of ethanoic acid in water. Ethanoic acid is weak because it ionises only partly in water, giving few $H^+$ ions.

**Q21.** Compound A has the formula $C_2H_6O$ and is found in alcoholic drinks. When A is heated with excess concentrated sulphuric acid at 443 K, it gives a gas…

**Answer:** A: ethanol. B: ethene. C: ethane.
$C_2H_5OH \rightarrow C_2H_4 + H_2O$ (conc. $H_2SO_4$, 443 K: dehydration)
$C_2H_4 + H_2 \rightarrow C_2H_6$ (nickel catalyst: addition/hydrogenation)